



The Platform Society

José van Dijck et al.

<https://doi.org/10.1093/oso/9780190889760.001.0001>

Published: 18 October 2018 **Available in print:** 29 November 2018

Online ISBN: 9780190889807

Print ISBN: 9780190889760

Search in this book

CHAPTER

7 Governing a Responsible Platform Society

José van

<https://doi.org/10.1093/oso/9780190889760.003.0008> Pages 137–162

Published: October 2018

Abstract

This chapter shifts the focus from the analytical and the descriptive to the normative and the reflective. A key issue is how public values can be forced upon the ecosystem's architecture—an architecture whose core is overwhelmingly controlled by (US) tech giants pushing economic values and corporate interests, often at the expense of a (European) focus on social values and collective interests. The mechanisms of datafication, commodification, and selection seem to afford tech companies unprecedented infrastructural, sectoral, and intersectoral powers. However, the ecosystem's architecture is adaptable to changing societal norms and awareness about potential harms. This book's search for underlying patterns and systemic mechanisms prompts a final reflection on the “what,” “how,” and “who” of governance: what kind of public values do we want to incorporate into the design of the platform society, how do we do that, and who is responsible for doing so?

Keywords: public value, democratic governance, platform society, value-centric design, information asymmetry, power unbalance

Subject: Economics, Digital Lifestyle, Media Studies

Collection: Oxford Scholarship Online

Introduction

The year 2017 was arguably a turning point in the development of the platform society; there were notable signs of an emerging opposition against the mounting power of large platform operators—and not just of the Big Five. Alphabet–Google, owner of YouTube, faced a strike from major advertisers, including the *Guardian*, Starbucks, and Walmart, that refused to have their ads placed alongside videos celebrating terrorism and disseminating fake news. The German government started to prepare legal measures against social networking sites that fail to take down hate speech, fake news, and defamatory content within 24 hours of its being posted. Twitter was reprimanded for not being able to comply with this voluntary rule. In the meantime, German competition authorities took Facebook to court for issuing blanket, incomprehensible consumer agreements that force users to cede all rights over their personal data to the company. And in anticipation of the General Data Protection Regulation (GDPR) directive—a ruling that will protect every European Union citizen from data abuse starting in May 2018—the European Commission considered whether social media sites’ harvesting of personal data breaches European Union antitrust rules. European antitrust officials fined Alphabet \$2.7 billion USD for unfairly favoring some of its own services (i.e., Google Shopping) over those of rivals, creating a lock-in for its users. In December 2017, the European Court ruled [↳] Uber to be a taxi service rather than a tech company, dealing the platform another severe setback after the denial of its license renewal in London and the public outcry over Uber’s harassment practices tolerated by its management earlier that year. Meanwhile in the United States, Facebook, Google, and Twitter executives were made to testify in front of the US House and Senate Intelligence Committee to probe their role in Russia’s meddling in the 2016 US elections. And finally, two Wall Street investors requested that Apple look into the health effects of its iPhone and iPad products and study whether their design may promote addiction, particularly among children.

A growing backlash against the big tech companies’ profound impact on all levels of society has drawn public attention to the intrinsic flaws of the global platform ecosystem (Manjoo 2017b). Each incident exemplifies the skirmishes between governments, market forces, and civil society actors to define the platform society as part of an ideological fight over public values and competing interests (Hutton 2017). In the first two chapters, we assessed how online platforms—with the Big Five squarely in the middle—are quickly positioning themselves as central to all sectors, penetrating private and public domains, affecting markets, and reshaping societies. Platform operators have gained not just economic power but also civic power, including “the power to enable collective action, the power to communicate news, and the power to influence people’s vote” (Moore 2016). Indeed, there are enormous economic benefits for the Big Five in maintaining an “in-between” status as “neutral connectors” while accumulating centralized informational power (Andrejevic 2013b). The next four chapters analyzed the disruption of urban transport and news as part of the market sector and health and education as predominantly part of the public sector. In shaping the online world, the micro-level of single platforms is inextricably intertwined with the meso-level of the platform ecosystem, which in turn feeds into the geopolitical level of global companies and (supra-)national government bodies.

In this last chapter, we want to shift our focus from the analytical and the descriptive to the normative and the reflective. We started out this book asking what role online platforms play in the organization of public values in American and western European societies. Public values and the common good often remain implicit or invisible, whereas they should be articulated, especially when they are contested or when they serve competing interests. A key issue is how public values can be forced upon the ecosystem’s architecture—an architecture whose core is overwhelmingly controlled by (US) tech giants pushing economic values and corporate interests, often at the expense of a (European) focus on social values and collective interests. The mechanisms of datafication, commodification, and selection seem to afford tech companies unprecedented infrastructural, sectoral, and *intersectoral* powers. [↳] However, the ecosystem’s architecture is adaptable to changing societal norms and awareness about potential harms. This book’s search for underlying patterns and systemic mechanisms prompts a final reflection on the “what,” “how,” and “who” of governance: what kind of public

values do we want to incorporate into the design of the platform society, how do we do that, and who is responsible for doing so?

As stated in the first chapter, creating public value for the common good should ideally be the shared responsibility of market, state, and civil society actors. The Big Five, as major shapers of the ecosystem, can rewrite the rules for democratic societies; but how can they act responsibly in a new global world order? The same question can be asked of civil society actors. What are the capabilities and constraints of individual citizens, nongovernmental organizations, and cooperatives in the platform ecosystem? Public institutions have a special function as guardians of social trust and a democratic commons. Finally, we will zoom in on the responsibilities of local, national, and supranational governments that are slowly waking up to the vast consequences of outsourcing governance to sociotechnical systems. Platformization requires an integral approach rather than just a sectoral one when it comes to defending consumers' and citizens' rights. Governments are, of course, regulators and law enforcers, but they can also be responsible users and developers of platforms. Governing the platform society is ultimately a geopolitical fight for influence and power; in the epilogue, we will return to the ideological clash between global powers in the digital age.

Toward a Value-Centric Design of the Platform Society

Societal orders are no longer just cemented in offline institutions and organizations, but they are increasingly interwoven with digital constellations. If societies want to create a platform structure that reflects and constructs a democratic order, they need to strive to implement public values and collective interests in the ecosystem's design. The current platform ecosystem is predicated on an architecture that is primarily staked in, and driven by, economic values and corporate interests; the more societal sectors this system penetrates, the more political power its key players accrue. Although the rapidly evolving ecosystem often appears as a self-driving force, it is not immune to change: technology shapes society as well as the other way around. The ideal platform society, which governance builds on homogeneous generic public values, obviously does not exist. Public values need to be articulated in each sector and in each context because they are both stakes and outcomes in political and ideological contests. Openly articulating and weighing public values at every level is imperative to revamping the ecosystem's design (Tura, Kutvonen, and Ritala 2017).

p. 140 In the first chapter, we introduced a range of public values, some pertaining to individual consumer rights, others to the process of keeping the Internet open and reliable. The *safety* of data traffic is, of course, a fundamental public issue that pops up in the news almost every day: data flows that can be tapped or online systems that can be hacked cause a fundamental mistrust in the usability of the infrastructure as a whole. *Privacy* is a value that has shown up throughout our discussions of data traffic via online channels. Without privacy, citizens can no longer be assured that they have control over their own lives; the right to autonomy and sovereignty of individuals, of course, should always be weighed against the interests of communities. *Transparency* is another pressing public value that applies not just to data flows and algorithms but also to business models and the ownership status of platforms. The import of *accuracy* often comes to the fore when discussing questions such as: is information correct? and how can we control its accuracy so we can assess the fairness of its interpretation? Such questions are important in any sector, but they can mean the difference between life and death in the health sector and between real and fake in the news sector.

More broadly, we have analyzed how societal values form the heart of debates over private gain versus public interests. In an ideal governance model, citizens would have the *power to control* their personal data and wield *democratic control* over what happens to collective data flows and repositories.¹ *Quality* and *accessibility* are important values when it comes to data-driven processes; as we have seen in the health and education sectors, these values are inherently subjective and therefore should be open to public debate. In the transportation sector, *comprehensiveness* and *affordability* came to the fore as contested public values. A society that holds on to

principles of *equality*, *inclusiveness*, and *fair treatment* without favoritism or discrimination needs to be able to judge automated decisions made by **algorithms that are now obscured from public scrutiny**. Two more general public values are *responsibility* and *accountability*. Public accountability suggests that it pertains to “good bookkeeping”—to be answerable for fiscal and social responsibility—but the term has moved far beyond its semantic origins and has come to signify “good governance,” both in the public and in the private sectors (Bovens 2009). Needless to say, the list of public values is not limited to the ones discussed in the preceding chapters.

While everyone agrees on the necessity of articulating public values, there are no universal guidelines or recipes for the ensuing negotiations; “fairness” in healthcare contexts means something different from “fairness” in self-driving cars or education. Articulating which values are contested by whom in which context may help reshape the current platform ecosystem in ways that make it more responsive to public concerns. Implementing public values in the ecosystem’s design is not as simple as it seems, partly due to the sector-agnostic logic inscribed in the architecture of ↪ infrastructural platforms as well as in the platform mechanisms. As we have shown in the preceding chapters, platforms’ techno-commercial strategies and user practices constantly co-evolve; and because of their **performative nature**, we should look into how platform mechanisms can also be deployed to *promote* public values and the common good. Let us probe a few scenarios to show how datafication, commodification, and selection might be “reverse-engineered” to inject public values into the system.

Datafication

The problems with datafication currently often relate to poor accessibility, unknown quality, indistinct ownership, and lack of accountability in the governance of data flows (Kitchin 2014). For the ecosystem of platforms to work effectively, it heavily relies on large quantities of data generated by a vast army of users who face a loss of control over their data after they have accepted a platform’s terms of service. Data ownership is a troubled concept, partly because it has not been defined legally or technically, partly because the disaggregation and reaggregation of data flows and the unbundling and rebundling of digital products takes place at the back end of platforms, out of users’ sight, and partly because the vastness and complexity of data capturing by infrastructural platforms frustrate data portability.² Conditions for access often get defined by the owners of data storage and computing facilities, which are generally handled by the very same companies that offer the software services and hardware devices on which users have come to rely. For instance, health data collected through fitness apps are routed through the infrastructural platforms that also channel connections between advertisers, news content, and users. Since data flows are near impossible to trace and because access to data and control of data quality are overwhelmingly in the hands of a few large corporations, there is little public accountability. Data have become a valuable resource constantly supplied by users, but they are treated neither as individual/personal property nor as a communal/collective resource. As long as data flows are considered proprietary assets, platform operators have the power to capture, store, and resell them without any recourse or compensation to the public.

And yet, datafication is a potentially powerful means to improve learning, enhance the circulation of news, optimize city traffic, and solve health problems of epidemic proportions. The availability and accessibility of (real-time) data shared through open standards could contribute significantly to the common good. In a seminal paper, the Organisation for Economic Co-Operation and Development (OECD) has argued how data streams in the transportation sector could be leveraged to improve traffic management and safety; however, platform operators “have ↪ concerns that sharing their data with the government would allow other competitors to access the same information and eliminate their competitive advantage, and hence the public ends up losing out” (Organisation for Economic Co-operation and Development 2016, 28). As we saw in chapter 4, cities could substantially benefit from the data gathered by Uber and other transport network services to optimize traffic flows. Public access to data could also help to make platforms more accountable; therefore,

some classes of data should be shared for the common good, provided that privacy is protected. In chapters 5 and 6 we elaborated on the importance of making data flows open, traceable, and reusable. In the domains of health and education, we are seeing a tendency toward privatization of user-generated data—a tendency that sits in tension with the need for more transparency and testing for accuracy. Values like privacy can be wielded both for and against the public interest in open data flows; that is why it is important to articulate them as clearly as possible so that citizens and authorities can balance competing interests and values at every level of negotiation.

There are probably as many advantages to datafication and algorithm-based decisions as there are concerns. Platform societies are quickly moving from rule-based algorithms to machine-learning algorithms, where human decision-making is increasingly outsourced to machines, deciding who gets loans, who gets insurance at what price, or who is granted parole. As we saw in chapter 5, there are great opportunities, for instance, in personalized healthcare that is more efficient; but the concerns are equally great—think of discrimination or preferential treatment in healthcare. Subtle biases or existing ideologies, prejudices, and inequalities can slip into the data sets but also in the processing or training of algorithms (Amoore and Piotukh 2015).³ Computing risks is always a combination of humans and machines, making it more important to develop reiterative processes for *judging* data sets and processing principles. Questions of accountability and responsibility get more complicated when systems become self-learning.

p. 143

Data should perhaps be treated like money—and not just to distinguish the data-rich from the data-poor. Data flows ought to be made transparent just as money flows are made “(ac)countable” in order for citizens to see who owns them, who can access them, and who may use them. Perhaps in the future it will become possible to install a public register, a cadastre of sorts, which keeps track of data flows and enables individuals to keep some data private while giving away some data under certain conditions (i.e., anonymity or reusability for nonprofit purposes only). Such an administrative ledger system should be open to all, and its registers should also be held accountable—perhaps operated by public agencies or at least controlled by registered data accountants.⁴ Accountability is not merely a technical matter; it strongly correlates with how institutions scaffold our social world (Reijers and Coeckelbergh 2016). Of course, states and supranational bodies like the European Union can regulate data flows to some extent, but there needs to be an agreed upon legislative framework that defines who can “own” which data flows—privately, corporately, or collectively. Several organizations are keen on developing new (regulatory or legal) frameworks for understanding data flows similar to flows of money and goods; needless to say, it requires a concerted effort to design an alternative framing for data ownership and access that resonates equally well with businesses, governments, and citizens.⁵ For these initiatives to take root, it is crucial for societies to develop an integral approach to data as a valuable currency accounted for in a trusted governance system.

In recent years, blockchain technology has been pushed as the new trust system where transactions can be managed and authorized through a decentralized ledger and records can be administered in a distributed fashion so that users can validate and count every interaction without having to ask permission from a central authority or organization (Mancini 2015). Blockchain proponents have argued that if we would only design an ideal technical system, the ideal society will inevitably follow: a society that no longer needs central organizations after every online transaction is made transparent. There are also a number of critics who have voiced serious concerns about the underlying assumptions of blockchain, from its technological determinist vision to its cyberlibertarian ideology (O'Dwyer 2016).⁶ It is beyond the scope of this book to discuss the pros and cons of blockchain; instead, the observation suffices that the very dispute about blockchain technology revolves around basic public values and competing interests: Who will benefit from blockchain? Is it accurate and safe? Who is accountable in case the system is hacked or proven insecure? Is it sustainable both in terms of energy efficiency and in terms of public trust? **Can institutions, citizens, market actors, and state actors collaborate and be involved in shaping this technology in line with democratically agreed upon values?** These

are the kinds of questions that need to be raised in order to assess the significance of *any* technology for a responsible platform society.

Commodification

p. 144 The mechanism of commodification that helps transform datafied activities into economic value could in principle also be deployed toward creating *public* value. If we look at the business, ownership, and governance models currently prevalent in the platform ecosystem, we have to conclude that a public value–based design is still a far-fetched ideal. Transparency is a condition for accountability, yet the online world is getting more opaque by the day. Platform economics, simply put, relies on connecting producers to consumers to advertisers to content; **the currencies involved in online transactions are money (or alternative crypto-currencies), data, attention, and user numbers.** A myriad of monetization schemes render the ecosystem notoriously intransparent: sometimes a transaction fee is paid by both sides (e.g., Airbnb, Uber); other times it happens through the creation of free(mium) content generated and paid for by users (e.g., YouTube, Coursera); platforms may capitalize on consumer attention through advertising (e.g., Facebook) or by selling data to third parties (e.g., 23andMe). An increase in users, similar to data and algorithms, intensifies a platform’s network effects, offering it a potential competitive advantage over other platforms (Prufer and Schottmuller 2017). Business models are often business secrets; how platforms compute the four currencies to create value is often hidden from the view of competitors and users. The distribution of ownership and governance models is equally nebulous. Competition from newcomers is effectively offset when the Big Five companies buy up platforms as soon as they start to scale or—not uncommon—when they copy successful features from competing platforms to enhance their own monetization strategy.⁷

Users have little or no insight into how platforms extract value from their online activities; the onus is always on the user to share more, feeding platforms’ infinite appetite for data. Some might argue that tech firms have turned into “banks,” exchanging data for money while packaging their products into complex services whose price dynamics have become incomprehensible to ordinary users. If Twitter or Facebook users have no clue how they get fake news items in their information streams, that is likely because they are ignorant about the business models’ underlying clickbait. With so much information asymmetry, one may argue that high-tech firms owe consumers an explanation of how a platform’s monetization scheme works. Their obscurity seriously hampers consumer’s comprehension of online services—a situation reminiscent of the period leading up to the 2008 banking crisis when abstruse subprime mortgages were sold to aspiring, unwitting homeowners. Like the banking sector, the high-tech sector should be held responsible for the legibility and legitimacy of their products and services. They are not “too big to fail.”

p. 145 And yet, transparency alone cannot make platform ecosystems more accountable. As Ananny and Crawford (2016) have convincingly argued, transparency coupled with accountability should always be contextualized: who is accountable for what and to whom? With regard to commodification, this requirement should apply not only to business models but also to ownership models and the governance of markets. And that is precisely the problem: datafication and commodification as mechanisms render fluid the distinction between for-profit and nonprofit, private and public, free and paid for, infrastructural and sectoral, complementors and connectors—and between global, national, and local markets. As we have shown in the case studies on health and education, corporate platforms often partner with nonprofits and public players, promising communal benefits yet obfuscating private interests. Transparency and accountability need to be embedded in formalized relationship structures in order to work properly, but what happens if online and offline realities no longer match?

The offline apparatuses for regulating physical spaces and fair markets appear to be premised on the very dichotomies that datafication and commodification mechanisms undermine. User activity, transferable data flows, and automated data profiles are new variables in the global platform economy—variables that are barely

accounted for in any kind of legal-regulatory framework. These variables apply across sectors, thus allowing for an unprecedented accumulation of digital resources and concentration of *intersectoral* power, facilitating not just vertical integration across infrastructural and sectoral platforms but also horizontal integration between platforms operating in different sectors. Algorithmic power accumulates exponentially when, for instance, health data combined with educational scores and retail purchasing data can be utilized to decide who should be offered discounts on their insurance. Datafication and commodification mechanisms across sectors and across nations beg for new taxonomies of governance which may render the ecosystem more transparent and accountable; such principles of ecosystem governance are needed so that they can be applied to specific contexts and situations. We will return to this issue in the last section.

Selection

When discussing the mechanism of selection, we have shown how the automated filtering, ranking, and personalization of data may stymie public values such as the need for accuracy and democratic control, particularly if they are intertwined with specific commodification strategies. In the news sector, as discussed in chapter 3, the mechanism of personalization can diminish the diversity in news consumption. Platform aggregators like Facebook and Google's News product thrive on the model of unbundling and rebundling content, targeting specific individuals who are already prepared to receive slanted news and advertising. Although they considerably boost audiences, news aggregators have stripped away news *content* from journalistic *contexts* and hence from professional curatorial systems. Promoting disinformation and creating filter bubbles are not in the public interest; as Facebook and Google have come to acknowledge, the platform mechanism of selection, **driven primarily by commercial rather than public values, may seriously hamper democratic processes and eventually run counter to these firms' corporate interests** (Napoli 2015).

p. 146 That does not mean, though, that selection mechanisms are intrinsically faulty; personalization and reputation rankings may also work to empower individuals. Personalizing one's medical profile may be very beneficial to an individual's health. And customized news filters may in principle lead to better-informed consumers. However, there is a precarious balance between individual, private benefits and collective, public interests: connective systems that are geared toward optimizing personal services tend to subjugate or preempt collective interests. If personalized health information leads to better diagnosis and treatment, everyone gains; if it leads to unacceptably high rates for disabled people's health insurance, the very same mechanism undermines a social contract based on solidarity and equality. By the same token, personalized news filters may suit a specific person's informational needs in terms of news and advertising; but if these automated filters lead to a sharply divided citizenship in which communities lack any kind of common ground, we seriously need to consider how they can be made responsive to the demands of a democratically organized society.

In sum, a value-centric design of the platform society requires a retooling of the current ecosystem by tilting its underpinning mechanisms toward *societal valorization*. Over the past three decades, states and communities have yielded influence and power to corporations; the classic division of responsibilities where governments look after public interests seems no longer viable. Public sectors, institutions, and governments are increasingly reliant upon the corporate platform ecosystem, so it is crucial to rethink the principles from which these systems are designed and how the mechanisms could be wielded to sustain democracy and guarantee an equal playing field. The question to ask first is: what kind of society do we want? What values need to inform the online systems that have become so insidious in organizing our societies? Once again, the outcome of this negotiation process is never stable for once and for all as it is part of a democratic process. However, ignoring the technological, economic, legal, and social complexity of the platform ecosystem and hoping the market will take care of itself because it is "naturally" tilted toward democratic balance would be a serious mistake (Jacobs and Mazzucato 2016). Therefore, we will now turn to respectively market, civil society, and state actors—recaptured in Figure 7.1—who share responsibility for the platform society and ask how each type of actor can contribute to a public value-centered design. Such design requires negotiation and co-creation between

multiple stakeholders from the very beginning, forcing them to organize themselves toward a common goal: how can various interests be balanced to create a responsible—that is, democratic, accountable, sustainable, fair—platform society cemented in public trust?

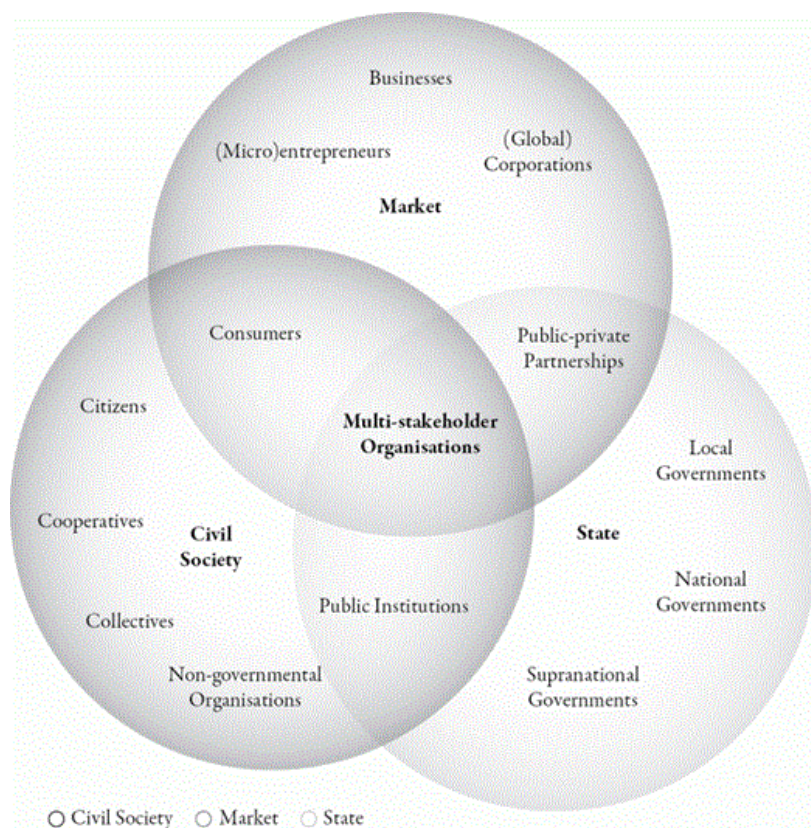


Figure 7.1. Schematic representation of actors from market, state, and civil society who shape the platform society; private and public actors cannot always be sharply distinguished and are partly overlapping. Collaborative efforts may be organized through multiple stakeholder organizations

(figure designed by Fernando van der Vlist).

p. 147 **Market Actors**

For the platform society to become democratic and sustainable, its various market actors—global tech companies, other (legacy) companies, micro-entrepreneurs, and consumers—need to put long-term public value creation over short-term economic gain. It is imperative to respect the position of all stakeholders, not just small and large businesses in the market sector but also civil society actors and governments. Responsible innovation starts with the inclusion of transparent values at the stage of design, acknowledging the various societal interests from the very onset (Taebi et al. 2014). This is not an easy task as values are contested and interests are fluid. What could various market actors do to add to a healthy democratic platform ecosystem?

Starting with the Big Five, we should reiterate the problem that they are wielding near-monopolistic power over essential infrastructural services while also deeply penetrating every market and public sector of society (Dolata 2017). The argument that platform operators want to be classified as generic technology companies and “connectors”—rather than as “utilities” or “sectoral services”—has long served as an excuse to operate outside the regulatory parameters that traditionally organize Western societies (Napoli and Caplan 2017). The Big Five’s powerful command over the platform ecosystem’s infrastructural core is increasingly challenged by (legacy) companies in various sectors as well as by critical consumers, journalists, and public

opinion.⁸ After several incidents in 2016 and 2017, with disturbing live videos being streamed on Facebook, racist videos appearing on YouTube, and hate speech swamping Twitter, social network companies started to deal more proactively with filtering this type of content. Platform operators who were reluctant to police or moderate content began to act more responsibly as a result of mounting pressure not only from US Congress and consumers but also from advertisers who threatened to withdraw their ads.

At this point, an important dilemma arises. Users increasingly want sectoral platforms to assume responsibility and comply with professional codes and regulations in specific sectors, such as news. At the same time, neither users nor legacy companies trust Facebook and Google as editors-in-chief of globally circulating online content as long as they deploy opaque selection criteria and business models which primarily serve the interests of their infrastructural platforms. Facebook's Mark Zuckerberg has suggested that social network platforms can be both technologically agnostic and politically neutral. Indeed, social media have lowered the barrier for political participation, but that is precisely what makes them useful to extremists and manipulation. Deciding what content is repugnant or which groups should be barred from the platform requires a willingness to accept moral standards paired off with a sense of public accountability and professional responsibility. Facebook talks about "community standards" and asks users to help them with flagging fake news and hate speech, but as long as Facebook does not publish the criteria by which its own human editors are supposed to weed out banned content, collaboration with multiple stakeholders will be difficult (Gillespie 2018). In sum, global selection criteria and national or local accountability demands may be profoundly at odds—a dilemma we also have seen in the health sector.

p. 149

Another problem that stems from power disparity is that users are simultaneously micro-entrepreneurs and consumers in the online marketplace, and yet they are not fully acknowledged as either one. Without users as co-producers of economic and cultural value, there would be no connective platforms. In spite of the fact that some platform owners close revenue-sharing deals with a few single productive users that draw huge crowds, for instance, a relatively small group of "influencers" on YouTube or Instagram, there is a growing resentment toward the tech giants' ↵ unfair capitalization of human creativity and (micro-)entrepreneurship (Taplin 2017). That resentment also pertains to big platforms' extractive relationships with micro-entrepreneurs in the service industry. Drivers for Uber and food deliverers for Deliveroo have no clue as to how a platform's reputation mechanism defines price dynamics and gig offers. They also appear to have little collective bargaining power to negotiate better working conditions and lower commissioners' fees. Users are not appropriately respected as consumers either. They are effectively "products" whose behavioral data contribute to profitable services which they subsequently consume; "free" services somehow seem to disengage operators from procuring regular customer protection rights. We can hardly speak of a "shared responsibility" if platform owners refuse to open up their algorithmic procedures (e.g., their ad rating systems, revenue models, reputation systems, or selection criteria) to micro-entrepreneurs and consumers; both complementors and consumers should be able to access them so that they can control their practices.

Theoretically, users can decide at any moment—individually or collectively—to opt out of Google, Facebook, Apple, Microsoft, and Amazon services; in practice, opting out is hardly an option for users who want to participate in society or who simply need to make a living. The more the ecosystem turns into a global connective utilities-like infrastructure, the more citizens become dependent on that system for their private, public, and professional activities. We cannot simply assume that individual consumers are savvy enough to fend for themselves when it comes to protecting their digital rights. The intricacies of data flows and algorithmic processing are simply too complex for users to understand the conditions to which they "agree" by checking a box. As some legal scholars have argued, individuals can hardly be expected to comprehend the full consequences of a single terms of service agreement when even trained lawyers admit they are unable to grasp them (Moerel and Prins 2016). So for protecting their rights as consumers in a situation of profound information asymmetry, users are largely dependent on governments and legislation.

Although some of the Big Five platforms have taken steps toward building social trust, statements of shared public responsibility have been (intentionally) vague. Mantras like “making the world more open and connected,” “accelerating science,” and “promoting health” conceal actual contributions to public value. A first step toward a responsible innovative design could be to open up the deployed mechanisms of datafication, commodification, and selection to public scrutiny so that they become transparent to users in specific contexts and countries. Open markets require public accountability from its major platform operators: openness about its technological and commercial platform mechanisms; openness about who drives communication and circulation of messages, particularly political and issue-based messages; openness about what its users—

p. 150

consumers actually buy into if they ignorantly sign terms of service. To understand private–public trade-offs, users need more clarification about sectoral vis-à-vis infrastructural power, about private versus public benefits. And, perhaps more urgently, users should be treated and respected as consumers as well as co-creators of economic and public value. The market can only regulate itself if the rules by which it is playing are transparent and open to all.

Secondly, platform owners and operators should start to realize that focusing on public values is a corollary to operating in Western democratic countries where checks and balances are anchored in institutional and legal frameworks. When Uber’s management came under fire for a score of scandals including harassment of its drivers in June 2017, customers and shareholders demanded that the company clean up its act and meet societal demands. The Uber case may count as an example of good working market self-regulation, but cleanup acts tend to come rather slowly, if they happen at all. In December 2017, Facebook announced it would start booking advertising revenue in countries where it is earned, instead of rerouting it to Ireland where the tax burden is known to be much lower. Facebook’s self-proclaimed insight that the company needs to provide more transparency to governments and greater visibility of locally generated sales and revenues to communities was more likely the result of a pending threat from European legislators to increase taxes on global digital companies. Abiding to rules for fair treatment of employees and reasonable taxation is part of a regulatory regime conventionally enforced by states and institutions. The question is whether the platform ecosystem is becoming less perceptive to governmental control now that the Big Five’s power and capital are beginning to exceed those of states. We will return to this issue in the epilogue.

Thirdly, tech companies often complain that (European) governments put up expensive, regulatory obstacles to enforce public values onto market sectors, hence thwarting innovation and economic progress. Argued along those lines, implementing a value-centred design in the platform society would be bad for business as well as for users. This kind of “stymied innovation” argument has been deployed frequently with regard to the reduction of fossil resources, until it became crystal clear that changing the energy paradigm could also have substantial economic benefits. Designing a public value-centered platform society should be considered not a liability but rather an asset: a loss of public trust is ultimately a loss of business value. This was prominently illustrated in March 2018 when Facebook lost almost 10% of its market value after a whistle blower revealed that Cambridge Analytica had been illegally allowed to tap into the private profiles of 50 million Facebook users—information that was subsequently used to influence individual citizens with political advertising. Both American and European regulators have started to investigate this case, but the bottom line is that Facebook experienced a major breach of public trust that cannot easily be mended. Platform operators that welcome and respect distinct contributions from government and civil society partners to revamp their design stand a better chance of succeeding in a sustainable platform society. This insight has increasingly dawned on some of the Big Five companies.⁹ Technological fixes alone cannot remedy complex political and societal problems; it takes a decentralized and multistakeholder approach to tackle such problems. Let us now turn to those other actors to see how they may approach this concerted effort.

p. 151

Civil Society Actors

Civil society actors are important partners in constructing a fair and democratic platform society; they first of all comprise engaged citizens but also nongovernment organizations (NGOs), cooperatives, and public institutions. The current ecosystem affords these actors a potent mix of empowerment and subjugation. But how can civil society actors help create and sustain a responsible platform society? What might be the role of engaged citizens, cooperatives, and public institutions in what Helbing et al. (2017, 12) have called a “new social contract, based on trust and cooperation, which sees citizens and customers not as obstacles or resources to be exploited, but as partners”?

Evidently, digital platforms can empower individual citizens to unite and to rally behind public advocacy efforts in order to influence democratic processes. Individual citizens are unlikely to influence big platforms’ policies, unless they manage to take their case all the way up to the European court.¹⁰ But individuals can muscle up collective power and become a force to be reckoned with because user metrics are one of the biggest currencies in the platform economy. For some time now, citizens have put their collective weight behind public concerns such as the right to privacy and access to one’s own data as well as the right to fair and accurate information. Collective user protests have sometimes managed to marshal power against the big tech companies, forcing platform operators to change their ways, even if only temporarily (Van Dijck 2013). Putting pressure on advertisers appears a more effective strategy because platforms like Facebook and Google are remarkably vulnerable to negative publicity.

p. 152 More generally, social media networks permit individuals to raise their voices for shared causes and connect at an unprecedented scale to pursue democratic goals (Bennett and Segerberg 2012; Castells 2012). A number of social movements that have evolved over time would have been unthinkable without the amenities of platform dynamics; as communication and organizing tools, social media platforms helped civil groups to act more as network nodes than as traditional social movement organizations.¹¹ Connective action has also been instrumental in ↵ the organization of local communities, as we have seen in the case of community-led transportation services. Since 2003, a substantial number of nonprofit user collectives have built platforms; some have transmuted into market ventures (e.g., Couchsurfing, TeachersPayTeachers), while other collaborations operating at the local level were gradually marginalized or simply disappeared. The problem that nonprofit collectives face in the platform society is a strong dependence on commercial infrastructural platforms with built-in mechanisms that are tilted toward global scaling and cross-sectoral data-sharing. These mechanisms appear squarely at odds with any desire to operate at a local or national level and to keep data traffic decentralized. Democratic control indeed seems a heavily contested public value in the current global ecosystem where local sovereignty or personal ownership over data flows is rarely an option offered by the dominant platforms.

The positive force of connective action combined with the ideal of citizens as organizers of collectivity has led some to advocate a system of “platform cooperativism” to offer an alternative to the global corporate ecosystem. As Trebor Scholz (2016) wonders, can citizens design, organize, and maintain an ecosystem of like-minded cooperatives in spite of (or outside of) the influential ecosystem led by the big tech companies? Why would self-employed taxi drivers, public schools, and neighborhood groups not build, run, and own connective platforms themselves? Scholz’s alternative does not just comprise a technological solution in the form of transparent data flows; his ideal of platform cooperativism also involves communal ownership and democratic governance, while different financing schemes and alternative legal rulings are indispensable to achieve his goal. In previous chapters, we have mentioned several interesting experiments with cooperative platforms, and there are many more.¹² While these efforts are viewed as viable alternatives by some, others have questioned their feasibility in the face of the current commercial ecosystem that capitalizes on a frictionless global user experience. Platform cooperativism may work well for small-scale communities, but it will not “spread”

automatically to other cities and countries, so it remains dubious whether they can present a countervailing power to the Big Five's dominant position (van Doorn 2017a).

p. 153 Indeed, there are very few examples of nonprofit and nongovernmental platforms that have been able to scale and establish a top-ranked position in the digital universe, while remaining true to their governance status. Arguably the most successful example to date of a nonprofit platform initiative on the Internet has been Wikipedia; ever since its creation in 2001, the "knowledge commons" has showed up consistently in the top ten of the most popular platforms, notwithstanding the fact that its laudable democratic ambitions are encapsulated within the corporate ecosystem.¹³ Wikipedia's most important feature is its remarkable system of collective governance that combines procedural transparency with a respect for neutrality, and it is this combination of pragmatism and idealism that has become the trademark of many "wikimedia commons" initiatives.¹⁴ A growing concern about the lack of public space in an online world has led to a call for a "technology commons" supported and developed by nonprofit collectives and researchers. Commons-based peer production, a term coined by law professor Yochai Benkler (2006), embraces the principles of collaboration and co-development, sharing of data and source codes, and developing alternative reward and recognition systems (P2P Value 2017). While such initiatives are laudable attempts to provide an alternative to the platform ecosystem, so far they have managed to leverage little power if they are not supported by either Big Tech or governments.

The need for *global* civil society actors, whether NGOs or commons initiatives, to defend and secure a noncommercial, actionable public space on the Internet has been forcefully claimed over the past few years. Organizations such as Creative Commons, started in 2001 by Lawrence Lessig, and Tim Berners Lee's Open Society Foundation have unremittingly supported a democratic, fair, and secure Internet where citizens can exercise democratic control over information and data flows. They have been promoting open source and open data initiatives in an Internet environment that is increasingly determined by commodified mechanisms and proprietary algorithms, going against detractors who argue that these initiatives do not make a dent in the corporate system (Wittel 2016). In the spring of 2017, Tim Berners Lee, speaking for the sustained goals of the Open Society Foundation, revealed his three biggest worries regarding the future of the Web as a tool to serve humanity: the loss of control over our personal data, the ease at which misinformation can spread on the Web, and the obscure ways in which political advertising campaigns are now targeted directly at users. Pointing out the urgency of keeping the Web an open space for everyone, controlled by democratic forces and founded on principles of fairness and equal access, he stresses the need for NGOs and nonprofits to collaborate with companies *and* governments to negotiate the ground rules and common principles of a platform society.

Civil society initiatives are indeed direly needed, and there has been no lack of initiatives, particularly in Europe. Many plans for developing open source software or peer-to-peer platforms for civil society goals have been supported by state subsidies, but hardly any of them manage to scale from project to product.¹⁵ Paradoxically, the foremost reason why these initiatives have a hard time succeeding may be *because* they have no centralistic, global business model and therefore do not fit the start-up logic of raising venture capital for scaling up. It is precisely this incongruency that may aggravate the power imbalance between market and civil society actors—an imbalance that may be remedied by collective action combined with collaboration with independent institutions and better regulation.

p. 154 Finally, institutions such as schools, hospitals, universities, libraries, statistics agencies, and public broadcast organizations, which act independently from governments *and* corporations, have an important role to play as autonomous guardians of public values. Particularly in western Europe, public institutions still have a relatively strong position vis-à-vis the private sector, whereas in the United States public funding for education, health, and media has gradually dwindled. It is no coincidence that dominant corporate platform operators try to weaken the power of institutions by bypassing their *systems of trust* that were traditionally anchored in professional routines to ensure common knowledge. Think, for instance, of independent researchers and peer-reviewed methodologies; statisticians and agreed upon methods of analysis; doctors who ground their

expertise in medical protocols, such as double-blind testing; and public media journalists whose editorial decisions and data policies need to be open to public scrutiny. Perhaps ironically, the corporate platform ecosystem that preferably bypasses public institutions and their “cumbersome” processes stands to profit significantly from data flows that are kept open so that they can be reused and repurposed.

For the past several years, institutions have actively pushed for open data and open online resources, particularly in the areas of health, science, and education. There is a broad consensus among researchers that open data and open science standards will expedite public accountability; opening up health research data can save lives, and making educational data open can help youngsters to learn. But in pursuing open data policies, public institutions as well as governments supporting openness have so far ignored the elephant in the room: the overwhelming majority of data are gathered *outside* of public institutions. Tech firms solicit user-generated data, which are subsequently processed by tech company’s analytics tools, hence turning vital public resources into proprietary assets. Increasingly, scientists have to purchase behavioral and real-time health data as well as analytics tools from tech companies—an expense that public research can ill afford. Without reciprocity, data-driven markets will never become a level playing field. A communal definition of “openness” including fair rules for all stakeholders has yet to emerge. As it stands now, “open,” like “sharing,” will become an empty, nonreciprocal term; businesses can profit from open data produced by public institutions, while data and knowledge generated by users but processed by corporations become proprietary.¹⁶ Once again, “open” and “public” are not the same things.

p. 155

Independent public institutions should be encouraged to develop *public* data systems that are open, accountable, and fair. However, they often lack the expertise and financial support needed to develop a truly public alternative, as we have illustrated in chapters 5 (health) and 6 (education). So, while businesses get access to taxpayer-funded “open” data sets at no cost, at the same time they are effectively removing public institutions from the marketplace because they weaken their position as producer and legitimate assessor of such data (Kitchin 2014, 61). When public data processing gets outsourced to the automated instruments owned and operated by high-tech firms, their measurements and interpretations are also withdrawn from the public eye, so in the long run, these institutions are effectively privatized. In this book, we have not touched upon national statistics agencies as trusted “accountants” of public information. Impressive statistical data sets processed and publicized by these institutions start to look tiny when compared to the data sets accumulated by high-tech and data companies. Google and Facebook arguably collect more valuable data about our economy or well-being every single day than official statistics bureaus may possibly generate in a year. Rather than relying on public statistics, whose tools were always open to being checked by other stakeholders, we are increasingly dependent on data analytics firms; their proprietary algorithms give them a competitive advantage, and they will not yield access to their measuring methods (Davies 2017).

Public institutions have always been anchors of contextualized accountability, for instance, in the form of professional codes and, more generally, as instruments of democratic control. They are in perpetual danger of becoming underfunded and understaffed. If schools, universities, or hospitals are cut off from innovation, they will become stagnant or obsolete. As it stands now, the global platform ecosystem threatens to undermine institutions’ status as independent providers of common knowledge and public services. The gap between public and private knowledge gets aggravated when experts’ salaries in the private tech sector are much higher than anywhere else. The destabilization of public institutions comes at a moment when they are already weakened by the withdrawal of government funding and deregulation, particularly in the United States where political outcomes are defined by special interests rather than by the broader public good (Fukuyama 2014). Conversely, if institutions are stimulated to become active (co-)developers of online platforms, with a clear eye for ethical standards and a mandate for creating public value, they not only enhance a level playing field but also generate economic value. Such stimulus and support will have to come primarily from governments.

Governments

p. 156 As argued earlier, good *governance* need not only come from *governments* but requires a joint effort by all actors. However, governments have historically been assigned the political power to run regulatory systems installed to protect citizens and consumers. It is up to governments to “develop coherent and publicly accountable methods for identifying, describing, and responding to systemic threats” (Julie Cohen 2016, 374). The term “government” is often taken as a synonym for “regulation”—a term that typically causes widespread allergic reactions in Silicon Valley. Like the banking sector, high-tech companies extensively lobby politicians in Washington and Brussels in favor of self-regulation and against government intervention. Government bodies operating at local, national, and supranational levels can respond to the emergence of multisided, global platform markets in a variety of ways (Just and Latzer 2017; Musiani 2013; Mancini 2015). Besides enforcing the regulation of platforms, they can shift to interfering in the governance of platforms by articulating their public value demands to developers. Moreover, governments are not just *regulators*; they are also *users* and *developers* of platforms. We will discuss each of these roles in more detail.

Governments as Regulators

The task of governments as regulators of digital platforms is traditionally compartmentalized, with each level of government assuming a limited regulatory range. *Local* governments often negotiate public values with single platforms in specific sectors, as we have seen in the case of Airbnb with the city of Amsterdam and with Uber wanting to enter a local taxi market. Each city in each European country fends for itself when it comes to negotiating public values with digital newcomers and legacy companies, even though more recently some cities have started to coordinate their negotiation tactics.¹⁷ This is both a blessing and a curse for platform companies. On the one hand, they profit from the absence of a single European market guaranteeing a level playing field; on the other hand, they have to deal with a costly variety of local and national legislations. While local authorities often lack the influence to remedy “systemic threats,” as Julie Cohen (2016) called them, they can certainly leverage their assigned democratic power to send strong signals to higher-level governments. Local governments can muster power by actively articulating the public value standards to which platforms have to comply. Remember how São Paulo proposed a comprehensive online platform strategy to the entire sector by articulating its value-based demands for fair, accessible, sustainable, and affordable public transport in the city (chapter 4).

p. 157 So far, few *national* governments have initiated comprehensive approaches toward platformization. Most western European states have left it to market players and local authorities to respond to what they call “market failure”—the inefficient allocation of goods and services. National governments have been rather slow to design sectoral strategies, often concentrating on just a few specific sectors, such as urban transport or hospitality. They have also singled out a few legislative frameworks, such as consumer rights or privacy standards for regulatory control. Partial approaches have resulted in some provisional remedies with limited results. Studies on the sharing economy increasingly point toward the benefits of national governments designing comprehensive platformization strategies to ensure a fair and level playing field.¹⁸ Such intervention can take several forms, including the regulation of algorithms, imposing oversight on data flows, and adjusting currently inadequate antitrust legislation (Sauerwein, Just, and Latzer 2015). In addition, national governments can and should take on generic disputes caused by platformization, such as labor relationships involving micro-entrepreneurs, as discussed in chapter 4.

As far as the European continent is concerned, the onus of regulatory accountability has come to rest with the member states in constant interaction with the European Commission. Regulation at the supranational level has proven to be most effective with regard to antitrust and privacy protection. Indeed, the 2.4 billion euro fine Commissioner Margrethe Vestager imposed on Google in 2017 and the effectuation of the GDPR in 2018 issued

serious warning signals and had significant impact.¹⁹ However, even if companies face big fines in the case of noncompliance, these single pieces of European regulation all but counter the core problems of the ecosystem—most profoundly because most (supra-)national legislation addresses one single concern at a time, such as privacy, competition, and antitrust issues; consumer protection; and business and taxation laws. What is missing is a comprehensive approach that tackles the widespread structural disparities of power, welfare, and economic opportunity in an online world—an approach that recognizes the interdependent nature of all these legal frameworks (Rahman 2017). Anchoring public values and the protection of the common good into the legislative blueprints of platform societies is a daunting project, where several hurdles need to be overcome.

The first difficulty is that most national legislatures simply *do not have a fitting vocabulary* to capture the socio-technical finesses of an evolving ecosystem of platforms that threatens to undermine many established societal arrangements. They also *lack a refined taxonomy* of techno-commercial mechanisms that can adequately delineate power relationships between various actors. Data flows, path dependency, sector-agnostic algorithms, vertical and horizontal lock-ins, active users as currency for value accretion, and search engine degradation (Stucke and Ezrachi 2016), for instance, are not part of a common legal discourse; and neither are “filter bubbles” or “personalization.” Crucial concepts of regulatory oversight are still moored in a predigital, prenetworked system of governance leveled at local and state-based societies that all but match a platformized reality (Khan 2017).²⁰ In most (supra-)national legislation, the vocabulary of antitrust law, drafted to promote fair competition and to prevent monopolies or harmful mergers and acquisitions, does not accommodate the new conditions of interoperability at a cross-sectoral level and at a global scale, for instance, when it comes to vertically integrated online ad services, aggregator news services, social media networks, and news content producers. As Martin Moore, founder of the Media Standards Trust in the United Kingdom, concisely puts it in his manifesto for a democratic platform society:

The tools currently available to democratic governments—including legislation, regulation and taxation—are not well suited to dealing with the issues raised by the tech giants. These organisations are very large and transnational, often work to a different economic model to other corporations, and work in a communications environment that is fundamentally different from their predecessors. Until we better understand and communicate the dilemmas they raise, and until the public become concerned about the potential—or actual—threats they represent, it will be difficult to respond effectively. (Moore 2016, 59)

Another problem is that enforcement agencies that are assigned to execute regulation are *insufficiently equipped* to administer even the most basic form of accountability. Regulatory fixes require detailed insights into how technology and business models work, how intricate platform mechanisms are deployed in relation to user practices, and how they impact social activities. For instance, if digital bots impersonating humans should be outlawed, legislators first need the knowledge to understand how they function in order to develop the legal instruments to bar them. The same holds for requiring platform companies to be transparent about who is funding online political advertising: as we argued in chapters 2 and 3, platform mechanisms are very difficult to control because they play out across infrastructural and sectoral platforms. Moreover, they pertain simultaneously to constantly changing algorithms, obscure business models, and incomprehensible terms of service. Governmental regulators need specialized digital teams equipped with combined technical and legal knowledge to probe these complex mechanisms.

A third problem facing regulators and their enforcement agencies is that tech companies are allowed to thrive in a vague, nondescript operating space *between* infrastructural and sectoral platforms as well as *across* (public and private) sectors; under the current regime, these accumulations of market power elude the radars of public scrutiny. When data flows cannot be confined to a particular sector such as health, sector-specific risks appear to become unmanageable because data can be endlessly repurposed beyond the sector in which they were originally produced. **So far, big tech companies have resisted any legislation that treats them like sector-**

specific companies, be it media, transportation, or health firms. By the same token, they defy any attempt to become regulated as infrastructural providers or utilities, even if there has been a growing choir of public voices pointing at the possibility of such ruling in the United States (Lynch 2017; Swire 2017).

The European Court ruling which classified Uber a “taxi company” in 2017 was considered a milestone, just as the Google fine was considered a major step toward the break up of tech monopolies. However, there is still a long way to go for European regulators who are the metaphorical soldiers equipped with hammers and nails facing a well-furnished digital army resourced by deep pockets. Transnational organizations like the OECD have pleaded for the introduction of sophisticated methods of data analysis into law enforcement, “in order to detect and prevent concerted practices in the digital markets” (Organisation for Economic Co-operation and Development 2016, point 85). Preventing such “concerted practices,” though, will take a concerted effort from regulators to coordinate what now seems like a patchwork of battles on many legal fronts: antitrust litigation, data storage and privacy laws, labor laws, national security, tax law, and more. While the negotiation to balance public and economic values in pursuit of a sustainable platform society should indeed be a shared responsibility, state actors have the ultimate duty to guard and protect public trust. For states to perform this role, they may need not see a government’s function solely as defensive but also as proactive; besides regulators, governments are also users and developers of platforms.

Governments as Users and Developers

Governments and publicly funded organizations function as exemplary users of platforms that should be held to the highest standards of transparency and accountability. **If government workers select corporate platforms to use in professional contexts, they are supposed to align their choices with reigning public values in their field.** In the current platform ecosystem, though, they often have no choice but to succumb to built-in mechanisms, triggering fundamental questions about conflicting values. For instance, can a public school require its students to use Facebook’s login or mandate the use of a Chrome notebook loaded with Google software as a condition to participate in educational activities, knowing that student data are shared with these companies?²¹ Police officers and law enforcement agencies are now routinely using Facebook, WhatsApp, YouTube, and Twitter to monitor individual and group behavior online. However, the selection and filtering mechanisms built into their architectures are not free of biases, which may in turn replicate or amplify implicit values and lead to discrimination or ethnic profiling (Meijer and Thaens 2013). Researchers and medical doctors increasingly close deals with big tech companies where patient data are exchanged for analytics services, without looking carefully into the long-term consequences of this trade-off.

p. 160 Public sector professionals and civil servants have a huge responsibility in handling platforms; they should at least understand the immediate and preferably long-term effects of platforms’ architecture on their public mission. Teachers, police officers, and doctors can be held accountable for their appropriate deployment of platforms and handling of data; as public professionals, they should be exemplary users. And yet, every integration with the global ecosystem elicits questions about whether data flows and algorithms can be accessed in order to control their efficacy. Therefore, governments should be more demanding as proactive users: if they insist on value-centric-designed platform services, companies will have no other choice than to provide them. It is important to define at the national level which public values platforms should deliver in each sector and across sectors so that services can be aligned with those demands.

Besides being exemplary and demanding users, governments can also be proactive as *developers* of platforms. In the libertarian discourse prevalent in Silicon Valley, “government as developer” is often associated with the pejorative term “big government” where “big” equals inefficient and cumbersome. And yet, in the United States, as well as in many European countries, national governments and their institutions build digital services that are crucial in the daily lives of citizens, such as online taxation systems, digital ID services, electronic submission and application services, patient health systems, etc. Even if the implementation of these

digital systems rarely happens without hiccups, part of the problem may have been a double-hearted approach toward the government's role as a central actor in the platform society. In a number of European countries, government institutions (tax authorities, social security agencies, census bureaus, cadasters, etc.) each developed their own information and communications technology (ICT) systems from scratch, oblivious to the fact that many of these systems might later need to become interoperable.

Very few states have so far developed the idea that governments can develop central plug-ins to be enforced on infrastructural or sectoral platforms. Think, for instance, of a calibrated digital taxi meter open to accountants that digital transport services should integrate in their platforms. Another example is identification services, an area that is now quickly monopolized by Facebook's and Google's login service. Estonia is often mentioned as an advanced example of a proactive government ready to shape its platform society. One single portal gives the citizens of "E-stonia" access to taxation services, identification, cadastral information, public libraries, personalized health information, public transportation services, and much more (Kalmus, Talves, and Pruulman-Vengerfeldt 2013). And in 2015, the United Kingdom initiated the government-as-a-platform approach to offer a comprehensible platform service developed as ICT modules made available through one online service point (GOV.UK 2015).

p. 161 And yet, as Tim O'Reilly (2011, 2017) has persuasively argued, if governments want to weigh in as actors, it is inevitable they first design a comprehensive platform approach centered on public values and collective goals. Collaborating with citizens' groups, governments can become *facilitators of organized trust*. For European governments, the need for a public value-centered, platform-based, and data-driven ecosystem is quite urgent.²² According to O'Reilly (2017, 173): "Just as companies like Google, Facebook, Apple, Amazon, and Microsoft build regulatory mechanisms to manage their platforms, government exists as a platform to ensure the success of our society, and that platform needs to be well regulated." Ideally, governments assign distinctive roles to public, private, and nonprofit platform operators, hence promoting a platform society with checks and balances. Governments can, for instance, mandate the use of a nationwide co-developed ID-verification system that is open to democratic control by various nongovernment stakeholders and which is subject to strict privacy and security rules. Countries like Estonia and Sweden have taken promising first steps toward such systems.²³

Taking this concept one step further, one could argue that governments, nonprofits, and corporations could become participants in multistakeholder collaborations—independent cooperations that develop decentralized yet interoperable systems which put public values at the center of their design (Cowhey and Aronson, 2017). Rather than leaving the regulation of single platforms and market sectors mostly to local authorities, states can guarantee a level playing field where actors are held to conform to democratically agreed upon public values. This will help to create platform solutions that move beyond weighing sectoral costs and benefits and inject long-term public values into the system: democratic values pertaining to information accuracy, sustainability goals, collective costs such as social security and insurance, as well as the accessibility and affordability of services. For instance, to remedy the insidious problem of misinformation in a platform society, an independent High-Level Group instated by the European Commission (2018, 5) proposes multi-stakeholder collaborations between civil society groups, public institutions like universities, corporate platforms, governments, and news organizations that might "safeguard the diversity and sustainability of the European news media ecosystem."

Considering governments as developers and as partners in multi-stakeholder cooperations requires a more comprehensive approach to the platform society, an approach that reaches beyond governments' common roles as regulators and exemplary users. Whereas some municipalities see the "disruption" of local economies by platforms as an inevitable form of privatization, other cities may see it as an opportunity to revamp the collective system. As public developers, governments can proactively steer the platform society to achieve a balance between market, state, and civil society actors. States, after all, have always been entrepreneurial, taking the lead ↪ in creating common infrastructures that ideally procure democratic values while generating

economic value (Mazzucato 2013; Jacobs and Mazzucato 2016). Just as the United Nations sustainability goals can lead to a healthy combination of economic and public value and the Paris climate accord was meant to slow down climate change while also stimulating innovation, a responsible platform society requires forward thinking by governments. A mix of government intervention, regulation, and stimulus measures may enable collaborative frameworks and multistakeholder organizations that are better calibrated toward anchoring trust and accountability (Sauerwein, Just, and Latzer 2015).

A digital world in which large corporations have both an overwhelming market presence and the leverage to influence political actors gives rise to highly unbalanced *polities*. For democracies to work in the age of platformization, they need the concerted effort of all actors—market, state, and civil society—to build a sustainable and trustworthy global platform ecosystem, a system that comes equipped with distributed responsibilities as well as with checks and balances. Indeed, the question *who* governs the platform society and *how* it should be governed based on *what* values is complex and multifaceted. Most of all, it is a geopolitical issue that cannot be handled as a national or even supranational cause; therefore, in the epilogue of this book, we will return briefly to the geopolitics of platform societies. Even though geopolitics is distinctly not the focus of our analytical exercise because it is highly speculative, we feel a need to sketch the contours of its dynamics because the future of platform societies cannot be viewed apart from global political contexts.

Notes

1. As Fukuyama (2016) observes, there is no clearly agreed upon definition of “governance,” just as there is no single definition of “accountability.” On the one hand, a group of scholars defines governance as “activities by traditional governments,” whereas another group extends this definition to actors outside of governments. We tend to accept the broader definition, even if we agree with Fukuyama that “it is not so clear how accountability works . . . in a world in which public policies are implemented by a host of shadowy networked actors operating in parallel (or perhaps even at cross-purposes) with one another” (99).
2. Users who voluntarily yield their data to Facebook by signing the terms of service may not know Facebook has a partnership with Axciom, one of the largest data processing firms in the world which leverages over 500 million profiles of people, each of which covers some 1,500 “data points” on average.
3. In March 2016, Microsoft revealed that an experiment in “conversational understanding” had shown how it took less than 24 hours of training on Twitter feeds to teach an “innocent” artificial intelligence chatbot (named Tay) to parrot the misogynistic and racist remarks dominant in the tweets and in responses from users.
4. Some scholars have coined the possibility of launching “(big) data accountants” (Coyne, Coyne, and Walker 2017). In the emerging platform world, independent controllers of data flows are just as indispensable as those guarding the transparency of money flows (Julie Cohen 2016).
5. The Open Data Institute (2016) in London is just one example of a nongovernment, nonprofit organization that is keen on developing transparent frameworks that help maximize the creation of public and economic value out of data flows while articulating fair conditions for ownership and accessibility.
6. Some critics argue that blockchain itself sprouts from an extreme libertarian view that dismisses the very role of states and governments and may lead to a disempowerment of citizens and even to the emergence of a global society without states and governments (Atzori 2015; Golumbia 2017).
7. In 2017, Facebook was accused of copying Stories, a feature on Snapchat that allows users to share user-generated video montages that will disappear after 24 hours. The feature had also been “tested” on Instagram six months earlier. European countries have ordered Facebook to stop sharing WhatsApp data (including phone numbers) with its other platforms; Facebook is also facing scrutiny over WhatsApp from the European Union’s data protection taskforce (Article 29 Working Party).
8. For instance, advertisers and publishers are trying to impact the unfair advantages of adtech platforms owned by Facebook, Google, and Axciom (a Facebook partner) by requiring transparency in advertising metrics; the ads.txt initiative is one such example. And some legacy publishers have started national publishing platforms in response to challenges posed by Facebook and Google, for instance, media companies like Bonnier and Schibsted which started their own journalism platforms.
9. In late 2017 and early 2018, Facebook demonstrated a willingness and ability to change—if only after being scrutinized by

- US Congress—its handling of political advertising practices and the involvement of state actors on its platform, showing that platforms are not fixed but perceptive to normative changes in society. In January 2018, Mark Zuckerberg announced a sweeping overhaul of its News Feed feature in order to prioritize people’s “most meaningful personal interactions” over passive content after users had said they felt swamped by outside posts from publishers and brands.
10. In 2013, Austrian law student Max Schrems filed his privacy complaint against Facebook with the Irish Data Protection Commissioner because Facebook has its European headquarters in Ireland. The Irish court referred the case to the European Court of Justice, which ruled in 2015 that it was not allowed for companies to transfer personal data from the European continent to the United States because privacy safeguards in this country are not sufficient—a landmark win for a European citizen.
 11. Public advocacy efforts, such as a successful disruption of the Anti-Counterfeiting Trade Agreement negotiation process, have illustrated the power of social media organizing in democratic decision-making processes (Losey 2014).
 12. Besides La’Zooz in the urban transport sector (chapter 3) and MyData in the health sector (chapter 6), there are a number of other noteworthy examples, such as The People’s Ride in Michigan and Yellow Cab Cooperative (California), which built and now run their own taxi apps and where drivers also own the company and share the proceeds. Enspirial is a European network of professionals and companies that promotes social entrepreneurship (Pazaitis, Kostakis, and Bauwens 2017).
 13. Even for a renowned platform like Wikipedia, known for the dedication of its contributors, it would have been impossible to scale without the support of companies such as Google (Van Dijck, 2013, chap. 7).
 14. In May 2017, Jimmy Wales announced that, in the light of the wave of fake news distributed by major commercial platforms, he wanted to start “Wiki-Tribune”—a news platform that brings together journalists and a community of volunteers; together, they want to run a service that produces fact-based articles that can be easily verified and improved—a kind of Wikipedia for news.
 15. Examples of these initiatives in Europe are D-CENT (2018), a “Europe-wide project developing the next generation of open-source, distributed, and privacy-aware tools for direct democracy and economic empowerment”; DECODE (2018) is another European project that provides privacy-aware tools that “keep personal information private or share it for the public good”; and CommonFare (2018) is a “means through which collaboration, experimentation, and solidarity can be expressed and spread to the whole Europe.” In The Netherlands, a nonprofit, open source–based initiative to create an authentication system is I Reveal My Attributes (IRMA 2017); the IRMA app assigns the authority to reveal personal attributes to individual users and is designed to provide privacy, security, and flexibility for the electronic ID infrastructure.
 16. As Rob Kitchin (2014) concludes in *The Data Revolution*, “If open data merely serve the interests of capital by opening public data, but keeping proprietary data locked behind pay walls and protected by intellectual property regimes . . . then they have failed to make society more democratic and open” (61).
 17. Just in the past years, European cities have started to coordinate their efforts to leverage more power in the negotiation with specific platform services, such as Airbnb and Uber; the city of Amsterdam, for instance, has joined forces with Barcelona, Paris, Lisbon, Vienna, Madrid, and Reykjavik to take up the regulator’s challenge.
 18. One such report was published by a group of researchers from Norway; in their *Power in the Sharing Economy*, Newlands, Lutz, and Fieseler (2017) argue for a more encompassing regulatory approach to the sharing economy. In The Netherlands, the Rathenau Instituut produced a comprehensive report on how to deal with sharing platforms (Frenken et al. 2017).
 19. The GDPR will take effect on May 24, 2018, and will cover all twenty-seven member states of the European Union. The most important tenets of this rule include limitations to the collection of data: data should be obtained with the knowledge and consent of the data subject; data should be relevant to the purposes for which they are to be used and should be accurate, complete, and kept up-to-date; and data should not be used for anything other than the original intention without again notifying the data subject (European Union 2018). At the time of finishing this book (January 2018), we can only observe how some of the Big Five companies, in anticipation of this law, organized intensive publicity campaigns to encourage users to set their privacy settings. Google, for instance, started to alert users on its various platforms but also in newspapers and magazines to (re)set their privacy settings.
 20. As legal scholar A. Khan (2017, 803) eloquently argues in the *Yale Law Journal*: “To revise antitrust law and competition policy for platform markets, we should be guided by two questions. First, does our legal framework capture the realities of how dominant firms acquire and exercise power in the internet economy? And second, what forms and degrees of power should the law identify as a threat to competition? Without considering these questions, we risk permitting the growth of powers that we oppose but fail to recognize.”
 21. As we argued in chapter 6, in the United States we see a growing number of public school districts save costs by closing deals with one of the tech companies, particularly Google and Facebook. In Sweden, though, the implementation of Google software in public schools has led to major discussions about data surveillance (Lindh and Nolin 2016).
 22. In a 2017 policy brief, the Dutch Central Planning Agency proposed a number of governmental measures to guarantee a fair and public value-centered platform society, including a permit system for platforms, rules to render selection

mechanisms transparent, the duty to mark and filter harmful (mis)information, the right to a verified account, and the right to recognizable political advertisements (Straathof, Van Veldhuizen, Bijlsma 2017).

23. In 2010, Sweden developed a mobile bank app, based on its efficient BankID system—developed by a consortium of leading banks and the government—which promoted digital innovation and lessened dependency on less trustworthy transnational ID systems such as those managed by Facebook and Apple (Andersson-Schwarz 2017). The Swedish system differs from the Estonian example because the development of its electronic ID system BankID is the result of a collaboration between the government and nationally operating banks.